

Query:

D3 PROPOSED PACKAGE INSERT	1. Kindly remove `BP` from `Bleomycin Sulphate for Injection BP` name. Note: It was found the finished product (bleomycin sulfate for injection BP)is NOT fulfills all BP specifications. Therefore is not allowed to claim `BP` on label or package insert. 2. Please amend 'dosage forms and packaging ' to 'dosage forms and packaging available'.
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Response: Response: We have removed BP from the finished product. Please find enclosed updated D3

For the use of a Registered Medical Practitioner or a Hospital or a Laboratory only OR for Specialist Use only

Bleomycin sulfate for injection 15 Units

BIXEL

Composition

Each vial contains:

Bleomycin Sulphate Ph. Eur. equivalent to Bleomycin.....15 units

As a sterile, freeze-dried powder for reconstitution

Dosage Form

Powder for injection

Product Description

White to off-white cake in glass vial, on reconstitution gives clear colourless solution.

Pharmacology

Pharmacodynamics

Bleomycin is a basic, water-soluble glycopeptide with cytotoxic activity. The mechanism of action of bleomycin is believed to involve single-strand scission of DNA, leading to inhibition of cell division, of growth and of DNA synthesis in tumour cells.

Apart from its antibacterial and antitumour properties, bleomycin is relatively free from biological activity. When injected intravenously it may have a histamine-like effect on blood pressure and may cause a rise in body temperature.

Pharmacokinetics

Bleomycin is administered parenterally. After intravenous (IV) administration of a bolus dose of 15×10^3 IU/m² body surface, peak concentrations of 1 to 10 IU are achieved in plasma. Following the intramuscular (IM) injection of 15×10^3 IU peak plasma concentrations of about 1 IU/ml have been reported. The peak plasma concentration is reached 30 minutes after an IM injection.

Continuous infusion of bleomycin 30×10^3 IU daily, for 4 to 5 days, resulted in an average steady state plasma concentration of 100-300 milli IU/ml. After IV injections of bleomycin in a dose of 15×10^3 IU/m² body surface, the area under the serum concentration curve is, on average, 300 milli IU x min x ml⁻¹.

Bleomycin is only bound to plasma proteins to a slight extent. Bleomycin is rapidly distributed in body tissues, with the highest concentrations in skin, lungs, peritoneum and lymph. Low concentrations are seen in the bone marrow. Bleomycin could not be detected in cerebrospinal fluid after intravenous injection.

Bleomycin appears to cross the placental barrier.

The mechanism for bio-transformation is not yet fully known. Inactivation takes place during enzymatic breakdown by bleomycin hydrolase, primarily in plasma, liver and other organs and, to a much lesser degree, in skin and lungs. When bleomycin was administered as an IV bolus injection in a dose of 15×10^3 IU/m² body surface, initial and terminal half-lives were 0.5 and 4 hours respectively. Given as a continuous intravenous infusion in a dose of 30×10^3 IU daily for 4 to 5 days bleomycin disappears from plasma with initial and terminal half-lives of about 1.3 hours and 9 hours respectively. About two thirds of the administered drug is excreted unchanged in the urine, probably by glomerular filtration. Approximately 50% is recovered in the urine in the 24 hours following an IV or IM injection. The rate of excretion, therefore, is highly influenced by renal function; concentrations in plasma are greatly elevated if usual doses are given to patients with renal impairment with only up to 20% excreted in 24 hours. Observations indicate that it is difficult to eliminate bleomycin from the body by dialysis.

Indications

Bleomycin is indicated for palliation and treatment adjuvant to surgery and radiation therapy of the following neoplasms:

- Squamous cell carcinoma of the skin, head and neck and oesophagus (primary indication)
- Squamous cell carcinoma of the larynx, penis and uterine cervix
- Squamous cell carcinoma of the bronchus (response infrequent)
- Choriocarcinoma and embryonal cell carcinoma of the testis
- Advanced Hodgkin's disease and other lymphomas
- Mycosis fungoides.

Note: Use of bleomycin after radiation therapy is less successful than use before radiation therapy.

Bleomycin is bone marrow sparing and may be used when other cytotoxic agents are contraindicated.

Dosage and Method of Administration

Routes of administration

Bleomycin may be given by the intramuscular, intravenous, subcutaneous or intra-arterial routes.

Note: Because of the possibility of an anaphylactoid reaction, lymphoma patients should receive test doses of between 1-5 units, for the first two treatments. If no acute allergic reaction occurs within 4-6 hours, the balance of the dose may be given. Thereafter the regular dosage schedule may be followed, if no reaction occurs.

Adult dose

Initial treatment (intramuscular, intravenous or subcutaneous administration)
Total doses of over 300,000 IU (300 units USP) should be given with great caution.

10,000 to 20,000 IU (10 - 20 units USP)/m² of body surface area given weekly or twice weekly.

Alternatively, give 15,000 IU (15 units USP) daily for 7 days followed by three weeks off-treatment and repeat twice so that a total dose of approximately 300,000 IU (300 units USP) is administered.

Improvement of lymphomas and testicular tumours is prompt ie. within two weeks while response by squamous cell cancers may take as long as three weeks.

A therapeutic response should be observed as the total dose approaches 150,000 IU (150 units USP), if this is not seen, consideration should be given to other therapy.

Note: When bleomycin is used in combination with other antineoplastic agents, pulmonary toxicities may occur at lower doses.

Intra-arterial administration

Intra-arterial infusion/perfusion is employed when increased drug concentrations at the cancer site are desired. The suggested dosage schedule is 30,000 to 60,000 IU (30-60 units USP) once or twice a week until the total recommended dose of 300,000 IU (300 units USP) is reached.

Repeat treatment

In patients for whom a course of bleomycin treatment provides an initial but incomplete response, a repeat course is suggested. Patients who show superficial improvement after one course e.g. in cases of squamous cell carcinoma, may benefit from a second course of treatment to prevent recurrence.

A repeat course may be commenced after a minimum of 3-4 weeks following completion of the first course, providing no sign of pulmonary toxicity have been observed. A total dose of 150,000 IU (150 units USP) for repeat treatment is recommended.

Paediatric dose

No information available.

Geriatric dose

Adult dose should be used with caution, particularly in patients over 70 years.

In impaired liver function

Use adult dose with caution.

In impaired renal function

As bleomycin is mostly excreted unchanged and as there is a high correlation between renal bleomycin clearance and creatinine clearance, impairment of function may require reduction in dosage and careful monitoring for toxicity. Dosage reductions of 40-75% have been recommended for patients with creatinine clearance values of ≤ 40 mL/min.

Directions for reconstitution -

For intramuscular or subcutaneous injection: dissolve the contents of the vial in 1-5mL of Sterile Water for Injection or Sodium Chloride Intravenous Infusion 0.9%.

For intravenous or intra arterial injection: dissolve the contents of the vial in 5-10mL of diluent and administer slowly over a period of 10 minutes.

Suitable diluents are Water for Injections, Sodium Chloride Intravenous Infusion 0.9%.

Reconstituted solutions containing 150 to 15,000 IU (0.15 - 15units USP)/mL bleomycin prepared using the recommended diluents remain stable for periods of at least 24 hours when stored in the dark, at temperatures of 2-8°C or 25°C. Solutions of bleomycin sulfate in Sodium Chloride Intravenous 0.9% stored in the dark at 2-8°C for 10 days were chemically stable. However, in order to reduce the possibility of microbiological contamination, reconstituted injections should be used as soon as practicable after preparation. If storage of the reconstituted solution is necessary, store at 2-8° C for no more than 24 hours. Any unused portions must be discarded in compliance with acceptable procedures for the disposal of anticancer drugs.

Contraindications

Bleomycin is contra-indicated in patients with acute pulmonary infection or greatly reduced lung function.

Patients who have previously had a hypersensitivity or idiosyncratic reaction to bleomycin

Warnings and Precautions

Patients undergoing treatment with bleomycin should have chest X-rays weekly. These should continue to be taken for up to 4 weeks after completion of the course. If breathlessness or infiltrates appear, not obviously attributable to

tumour or to co-existent lung disease, administration of the drug must be stopped immediately and patients should be treated with a corticosteroid and a broad spectrum antibiotic. High oxygen concentrations should be used with caution in these cases.

Lung function tests which use 100% oxygen should not be used in patients who have been treated with Bleomycin. Lung function tests using less than 21% oxygen are recommended as an alternative.

When bleomycin has been administered pre-operatively, reduced oxygen concentrations should be used during operation and post operatively.

Patients treated previously or concurrently with radiation to the chest may develop more frequent or severe toxicity. Dosage adjustment may be necessary.

Bleomycin should be used with caution in patients with significant renal impairment as clearance may be reduced and toxicity increased. (see Dosage & Method of Administration)

Bleomycin should be used with caution in patients with severe heart disease.

Drug Interactions

When bleomycin is used as one of the drugs in multiple chemotherapy regimes the toxicity of bleomycin should be borne in mind in the selection and dosage of drugs with similar toxic potential. Patients should be closely monitored for signs of pulmonary toxicity.

Concurrent use of antineoplastic agents may result in increased bleomycin toxicity, or in occurrence of pulmonary toxicity at lower doses of bleomycin. The addition of other cytotoxic drugs can necessitate changes and dose alterations.

Cisplatin-induced renal function impairment may result in delayed clearance and bleomycin toxicity even at low doses. Increased incidence of pulmonary toxicity has been noted when bleomycin is given with cisplatin. Dosage reduction may be required. Patients should be closely monitored for signs of pulmonary toxicity.

Previous or concurrent radiotherapy to the chest is an important factor in increasing the incidence and severity of lung toxicity.

Because of bleomycin's sensitisation of lung tissue, patients who have received bleomycin pre-operatively are at greater risk of developing pulmonary toxicity when oxygen is administered at surgery and a reduction in inspired oxygen concentration during operation and post-operatively is recommended (see Warning & Precautions). Dosage reduction may be necessary. Anaesthetics, general, and oxygen

Use of general anaesthetics in patients previously treated with bleomycin may result in rapid pulmonary deterioration, since bleomycin causes sensitisation of lung tissue to oxygen.

In patients treated for testicular cancer with a combination of bleomycin and vinca alkaloids a syndrome has been reported corresponding to morbus raynaud, ischaemia which can lead to necrosis of peripheral parts of the body (fingers, toes, nose tip).

The following clinical incompatibilities have been noted: Concomitant use of bleomycin with clozapine should be avoided due to an increased risk of agranulocytosis.

Granulocyte colony stimulating factor (G-CSF): It has been suggested that concomitant administration of G-CSF and bleomycin may increase the risk of bleomycin-induced pulmonary toxicity, especially at higher doses, although this has not been confirmed in clinical trials. If G-CSF is added to bleomycin-containing treatment regimens, patients should be closely observed for signs of pulmonary toxicity

Digoxin: Serum levels of digoxin may be reduced and its actions may be decreased. It is thought that drug-induced alterations of the intestinal mucosa may be involved in the reduced GI absorption.

Phenytoin: Serum concentrations of phenytoin may be decreased due to decreased absorption or increased metabolism of Phenytoin.

Incompatibility

Bleomycin solution should not be mixed with solutions of essential amino acids, riboflavine, ascorbic acid, dexamethasone, aminophylline or frusemide

Pregnancy and Lactation

Bleomycin should not normally be administered to patients who are pregnant or to mothers who are breast-feeding.

Pregnancy:

Bleomycin has caused, is suspected to have caused or may be expected to cause, an increase incidence of human fetal malformations or irreversible damage. It may also have adverse pharmacological effects.

Lactation:

It is not known whether bleomycin is excreted in breast milk. Due to the potential for serious adverse effects in infants, it is recommended that breastfeeding is discontinued prior to administration of bleomycin sulfate to the mother.

Undesirable Effects

The most frequently observed adverse reactions receiving bleomycin were pulmonary manifestations such as interstitial pneumonia or pulmonary fibrosis, sclerosis of skin, pigmentation, fever and rigors, alopecia, anorexia and weight decrease, general malaise, nausea and vomiting, stomatitis and nail changes.

Haematologic:

Common Haemorrhage

Uncommon Leukopenia

Nervous:

Common Headache

Uncommon Dizziness

Respiratory:

Very common Interstitial pneumonia and pulmonary fibrosis

Gastrointestinal:

Very common Anorexia, weight decrease, nausea and vomiting and stomatitis

Common Angular stomatitis

Uncommon Diarrhoea

Hepatic:

Uncommon Hepatic disturbances

Skin:

Very common Hypertrophy of the skin, pigmentation, alopecia and deformation and discolouration of the nail

Common Rash, urticaria and erythroderma associated with fever

Renal and Urinary:

Uncommon Oliguria, micturition pain, polyuria and feeling of residual urine

General Disorders and Administration Site Conditions:

Very common Fever, rigors and malaise

Uncommon Pain at the tumour site

Hypertrophy of the venous wall and narrowing of the venous lumen when given by intravenous injection

Induration when given by intramuscular or local injection

Like most cytotoxic agents bleomycin can give rise to both immediate and to delayed toxic effects. The most immediate effect is fever on the day of injection. Anorexia, tiredness or nausea also may occur. Pain at the injection site or in the region of the tumour has occasionally been reported, and other rare adverse effects are hypotension and local thrombophlebitis after intravenous administration.

The majority of patients who receive a full course of bleomycin develop lesions of the skin or oral mucosa. Induration, hyperkeratosis, reddening, tenderness and swelling of the tips of the fingers, ridging of the nails, bulla formation over pressure points such as elbows, loss of hair and stomatitis are rarely serious and usually disappear soon after completion of the course.

The most serious delayed effect is interstitial pneumonia, which may develop during, or occasionally after, a course of treatment. This condition may sometimes develop into fatal pulmonary fibrosis, although such an occurrence is rare at recommended doses. Previous or concurrent radiotherapy to the chest is an important factor in increasing the incidence and severity of lung toxicity.

A few cases of acute fulminant reactions with hyperpyrexia and cardiorespiratory collapse have been observed after intravenous injections of doses higher than those recommended. Hypotension, hyperpyrexia and drug-related deaths have been reported rarely following intra-cavitary instillation of bleomycin.

During post-marketing surveillance the following events have been reported: sepsis, pancytopenia, thrombocytopenia, anaemia, neutropenia, chest pain, myocardial infarction, raynaud's syndrome, embolism, thrombosis and digital ischaemia.

Overdose

The acute reaction to an overdosage of bleomycin would probably include hypotension, fever, rapid pulse and general symptoms of shock.

Treatment is purely symptomatic. In the event of respiratory complications the patient should be treated with a corticosteroid and a broad-spectrum antibiotic. There is no specific antidote to bleomycin.

Shelf-Life

2 years

Storage condition

Store at 2°C to 8°C. Refrigerate. Do not Freeze.

After reconstitution, Store at 2°C to 8°C or 25°C for no more than 24 hours.

Dosage form and packaging available

Carton containing one vial having fill value of 6 ml of BIXEL (Bleomycin Sulphate for Injection 15 units)

Product Registration Holder

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